

Supplementary Reading Material: On Dual Spaces

- 阅读提示: 无限维线性空间的 对偶空间 是一个含糊的概念. 通常的教材上有如下三类处理方式:
 1. (右倾) 仅考虑可构造的 对偶空间 , 例如 $(\mathbb{R}[x])^* \simeq \mathbb{R}[[x]]$, 但对一般情形避而不谈;
 2. (折中) 明面上不使用甚至不承认 选择公理 , 但在背地里默认了一些反直觉的公理: 例如默认非零空间存在非零的 对偶空间 , 但从不给出证明;
 3. (左倾) 直接使用 选择公理 .

有探索欲的读者不会接受右倾的处理方式, 而 选择公理 不在教学范围内. 本节习题将折中派的做法进行公理化, 用一条表述简单的额外公理为教材打上恰当的补丁. 我们最终承认通常公理与额外的 HB 公理.

- (通常公理). $\underline{\text{ZF}}$ 与 $\underline{\text{DC}}$, 也就是矩阵论和数学分析承认的东西. 特别地, $\underline{\text{DC}}$ 用于证明可数个可数集的并仍可数, 以及无限维线性空间存在无穷大的线性无关组, 其严格独立于 $\underline{\text{ZF}}$.
- (HB 公理). 对任意的子空间 $U \hookrightarrow V$, 映射 $f: U \rightarrow W$ 的定义域总能被扩大至 V . 换言之, 总能找到 $\tilde{f}: V \rightarrow W$ 使得 $\tilde{f}|_U = f$.

可以发现, [张贤科](#)等教材的公理体系严格等价于通常公理 + HB 公理.

若不承认 HB, 我们无法排除 Exotic Examples (见习题) 的存在性. 可使用这些例子自查是否伪证了某些命题, 类似地, 有限域的重要作用也是自查伪证与否.

习题目录: 有限维 对偶空间 一瞥, 不承认 HB 的后果, 承认 HB 的好处, 对偶空间的 Galois 对应 , 对偶空间的所有基本结论 .

Finite-Dimensional Vector Spaces and Dual Spaces

Definition. Let V be a vector space over a field $\mathbb{F}$. The dual space is defined as

$$V^* := \text{Hom}_{\mathbb{F}}(V, \mathbb{F}).$$

Example. For $\dim V = n < \infty$, with basis $u_{i \ 1 \leq i \leq n}$, the dual space V^* is also n -dimensional with basis (dual basis)

$$\{f_i\}_{1 \leq i \leq n}, \quad f_i: V \rightarrow \mathbb{F}, \quad u_j \mapsto \delta_{i,j}.$$

From the perspective of bilinear forms, the pairing

$$V \times V^* \rightarrow \mathbb{F}, \quad (u, f) \mapsto f(u)$$

has matrix form $I_{n \times n}$ under the bases $\{u_i\}_{1 \leq i \leq n}$ and $\{f_i\}_{1 \leq i \leq n}$.

Note. The isomorphism $V \simeq V^*$ is artificial. An informal but illustrative analogy: assume V has unit kilometre (km), then V^* has unit km^{-1} . The linear isomorphism $V \simeq V^*$ is never natural.

Example. The canonical linear map $\eta_V: V \rightarrow V^{**}$ is a natural isomorphism.

- The term "natural" signifies that for any linear map $f : U \rightarrow V$, the compositions $U \xrightarrow{f} V \xrightarrow{\eta_V} V^{**}$ and $U \xrightarrow{\eta_U} U^{**} \xrightarrow{f^{**}} V^{**}$ coincide.

Exotic Examples in Linear Algebra Without the Axiom of Choice

Example. ($\dim V = \infty$, yet $V^* = 0$). If the axiom of choice is not assumed, then there do exist infinite-dimensional vector spaces V with dual space $V^* = 0$. The [example](#) by H. Läuchli presents an infinite-dimensional vector space V whose proper subspaces are all finite-dimensional.

Example. (When V^* is strictly larger than V , yet $V \simeq V^{**}$). In ZF set theory with dependent choice, it takes effort to verify $\mathbb{R}[[x]]^* \simeq \mathbb{R}[x]$. However, $\mathbb{R}[[x]]$ does not possess a countable basis.

Proposition. The canonical linear map $V \rightarrow V^{**}$ is injective if and only if V is a subspace of some product space $\prod_X \mathbb{F}$. For instance, $\mathbb{F}[x]$ and $\mathbb{F}[[x]]$ both embed into their double dual spaces.

Proof. See [this article](#).

Axiom of Extension

Axiom. (Axiom of Extension, also known as the Hahn–Banach Axiom (HB)).

- For any linear subspace $U \subseteq V$, every linear map $f : U \rightarrow W$ extends to a map $\tilde{f} : V \rightarrow W$ such that $\tilde{f}|_U = f$.

Remark. The Hahn–Banach Axiom is independent of the usual axioms (ZF + DC), which are assumed in the study of tensor products of infinite-dimensional vector spaces.

Exercise. The Hahn–Banach Axiom is equivalent to the assertion that $\text{Hom}_{\mathbb{F}}(-, W)$ always maps injections to surjections.

对 $U \hookrightarrow V$, 映射 (π, W) 是满的, 当且仅当一切 $U \rightarrow W$ 通过 $U \hookrightarrow V$ 分解. 等价地, 这说明一切子空间 U 出发的映射可以提升至全空间 V .

Exercise. By HB, for any $v_1, v_2 \in V$ with $v_1 \neq v_2$, there exists $f \in V^*$ such that $f(v_1) \neq f(v_2)$. Hence, the canonical linear map $V \rightarrow V^{**}$ is always injective.

不妨设 $v_2 = 0$, 将映射 $\text{span}(v_1) \simeq \mathbb{F}$ 的定义域提升至全空间 V 即可.

作为推论, 记 $V \rightarrow V^{**}$ 为 $u \mapsto \delta_u$. 依定义, $\delta_u \neq \delta_v$ 的充要条件是存在 f 使得 $f(u) \neq f(v)$. 证毕.

Exercise. By HB, show that for $U \hookrightarrow V$, the following map is an isomorphism

$$\frac{V^{**}}{U^{**}} \rightarrow (V/U)^{**}, \quad [x]_{U^{**}} \mapsto \pi^{**}(x),$$

where π is the natural quotient map $V \twoheadrightarrow V/U$.

依照 $(-, \mathbb{F})$ 的性质, $(V/U)^* \xrightarrow{\pi^*} V^*$ 是 $V^* \xrightarrow{i^*} U^*$ 的 kernel. 依照 HB, $V^* \xrightarrow{i^*} U^*$ 是满的. 这一步骤示意作

$$[U \xhookrightarrow{i} V \twoheadrightarrow{\pi} (V/U)] \implies [(V/U)^* \xhookrightarrow{\pi^*} V^* \xrightarrow{i^*} U^*].$$

重复以上步骤, 得二次对偶的情形, 即 $V^* \xrightarrow{\pi^{**}} (V/U)^{**}$ 是满射, kernel 通过 $U^{**} \hookrightarrow V^{**}$ 实现.

Dual Spaces and Galois Correspondence

Definition. (Kernels and Annulateurs). For convenience, let U_i denote subspaces of V , and B_i subspaces of V^* .

- $\text{ann } U := \{f \in V^* \mid f(U) = 0\}$, a subspace of V^* ;
- $\ker B := \{u \in V \mid f(u) = 0 \ (\forall f \in B)\} = \bigcap_{f \in B} \ker f$, a subspace of V .

These notions are defined for subsets in general. By the closure property:

- $\text{ann}(X) = \text{ann}(\text{span}(X))$, for $X \subseteq V$;
- $\ker(S) = \ker(\text{span}(S))$, for $S \subseteq V^*$.

Remark. $\ker f$ and $\ker f$ are identical.

Exercise. Show that for any finite-dimensional vector space V , the map $\text{ann} : \text{Subspaces of } V \rightarrow \text{Subspaces of } V^*$ is a bijection with inverse map $\ker$.

答: 对任意子空间 $B \subseteq V^*$, 取 B 的基 $\{f_i\}_{i=1}^r$, 并扩充至 V^* 的基 $\{f_i\}_{i=1}^n$, 记对偶基 $\{r_i\}_{i=1}^n$. 此时 $\ker B = \text{span}(\{e_i\}_{i>r})$. 相应地, $\text{ann}(\ker B) = \text{span}(\{f_i\}_{i=1}^r) = B$. 另一方向同理.

Proposition. One has $\text{ann}(\ker B) \supseteq B$. Equality does not generally hold.

Proof. The formula $\text{ann}(\ker B) \supseteq B$ involves logic more than algebra. For a counterexample, take V as the space of continuous functions over $\mathbb{R}$, and $B = \text{span}(\{\delta_r\}_{r \in \mathbb{Q}})$.

Exercise. (Galois Connection)

1. $\text{ann}(\ker B) \supseteq B$,

证明略.

2. $\ker(\text{ann } U) \supseteq U$,

证明略.

3. $\text{ann}(\ker(\text{ann } B)) = \text{ann}(B)$,

由前两问, $\text{ann}(\ker(\text{ann } B)) \subseteq \text{ann}(B) \subseteq \text{ann}(\ker(\text{ann } B))$.

4. $\ker(\text{ann}(\ker B)) = \ker B$.

同上.

The pair $(\ker, \text{ann})$ defines a Galois correspondence.

Example. For a set map $f : X \rightarrow Y$, the image and preimage maps (f_*, f^*) form a Galois connection between $\text{Subset}(X)$ and $\text{Subset}(Y)$.

这是 17-Mar 的作业题.

Definition. Let $Op(\mathbb{R}^2)$ denote the set of open subsets of $\mathbb{R}^2$, and $Cl(\mathbb{R}^2)$ the set of closed subsets. Define:

1. $k : Op(\mathbb{R}^2) \rightarrow Cl(\mathbb{R}^2), \quad U \mapsto \overline{U}$ (closure);
2. $i : Cl(\mathbb{R}^2) \rightarrow Op(\mathbb{R}^2), \quad K \mapsto K^\circ$ (interior).

Exercise. Show that $ikik = ik$ and $kiki = ki$; hence, (k, i) form a Galois connection.

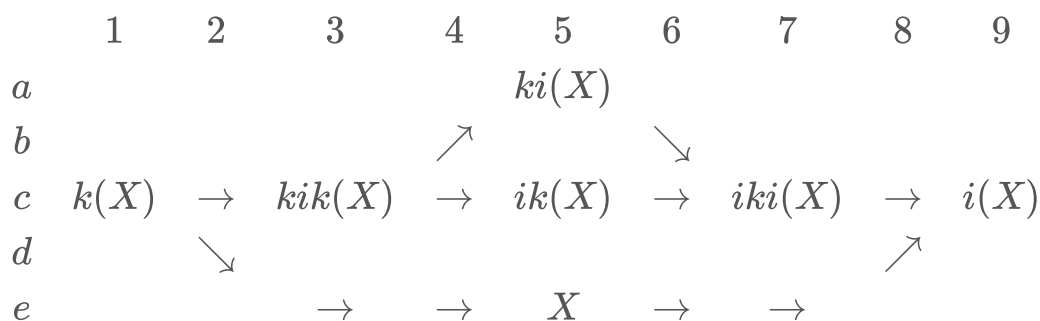
对任意集合 X , 都有

$$ik(X) = iik(X) \subseteq ikik(X) \subseteq ikkk(X) = ik(X).$$

$kiki = ki$ 证明类似. 对开集 $kik = k$ 是显然的; 对闭集, $iki = i$ 是显然的.

Exercise. Given $X \subseteq \mathbb{R}^2$, show that at most seven distinct sets (including X) can be obtained by applying k and i repeatedly.

依照运算关系 $ii = i, kk = k, ikik = ik, kiki = ki$, 至多有 $\{X, i(X), k(X), ik(X), ki(X), iki(X), kik(X)\}$ 两两不同. 考虑排序



为表述方便, 以下给出 $\mathbb{R}$ 上的反例.

- (2d 不取等) 考虑单点.
- (4b 不取等) 考虑 $[a, b] \cap \mathbb{Q}$.
- (4c 不取等) 考虑 $[c, d]$.
- (6b 不取等) 考虑 $[c, d]$.
- (6c 不取等) 考虑 $[a, b] \cap \mathbb{Q}$.
- (8c 不取等) 考虑 $[x, y) \cup (y, z]$.
- (3e 不取等) 考虑 $[a, b] \cap \mathbb{Q}$.
- (7e 不取等) 考虑 $[c, d]$.

令 $a < b < c < d < x < y < z$ 即可.

Exercise. Show that one can obtain at most 14 distinct sets (including X itself) by applying i and $(-)^c$ repeatedly.

记 $(-)^c$ 为运算 c . 注意到 $ic = kc$, 以及 $kc = ci$. 对任意 $\{k, i, c\}$ 的组合式, 可以将所有 c 移至开头, 使得式子中有且仅有 0 或 1 个 c . 往后同上.

The Yoga of Dual Spaces

Theorem. In the following table, FD denotes finite-dimensional cases, G denotes general cases, and HB refers to cases where the Hahn-Banach Axiom is assumed.

For a linear map $\varphi : U \rightarrow V$, define the dual map as

$$\varphi^* : V^* \rightarrow U^*, \quad [V \rightarrow \mathbb{F}, v \mapsto f(v)] \mapsto [U \rightarrow \mathbb{F}, u \mapsto f(\varphi(u))].$$

No.	Formula	FD	G	HB
G1	$\text{ann}(\ker B) \stackrel{?}{=} B$	=	$\supseteq$	$\supseteq$
G2	$\ker(\text{ann } V) \stackrel{?}{=} V$	=	$\supset$	=
E1	$\text{ann}(V_1 + V_2) \stackrel{?}{=} \text{ann}(V_1) \cap \text{ann}(V_2)$	=	=	=
E2	$\text{ann}(V_1 \cap V_2) \stackrel{?}{=} \text{ann}(V_1) + \text{ann}(V_2)$	=	$\supseteq$	=
E3	$\ker(B_1 + B_2) \stackrel{?}{=} \ker B_1 \cap \ker B_2$	=	=	=
E4	$\ker(B_1 \cap B_2) = \ker B_1 + \ker B_2$	=	$\supseteq$	$\supseteq$
Z1	$\ker \varphi^* \stackrel{?}{=} \text{ann}(\text{im } \varphi)$	=	=	=
Z2	$\ker \varphi \stackrel{?}{=} \ker(\text{im } \varphi^*)$	=	$\subseteq$	=
Z3	$\text{im } \varphi^* \stackrel{?}{=} \text{ann}(\ker \varphi)$	=	$\subseteq$	=
Z3'	$\text{im } \pi^* \stackrel{?}{=} \text{ann}(\ker \pi)$, with π surj.	=	=	=
Z4	$\text{im } \varphi = \ker(\ker \varphi^*)$	=	$\subseteq$	=
Z5	If f is surj., then f^* is inj.	true	true	true
Z6	If f is inj., then f^* is surj.	true	NaN	true
L1	$S^* \stackrel{?}{\rightarrow} V^* / \text{ann}(S)$	$\simeq$	$\hookleftarrow$	$\simeq$

No.	Formula	FD	G	HB
L2	$(V/S)^* \xrightarrow{?} \text{ann}(S)$	$\simeq$	$\simeq$	$\simeq$

(G1) FD, G 见上. G 反例: V 为 $\mathbb{R}$ 上连续函数, $B = \text{span}(\{\delta_r\}_{r \in \mathbb{Q}})$.

(G2) FD, G 见上. 若有 HB, 任取 $v \in \ker(\text{ann } V) \setminus V$. 定义

$$\varphi : \text{span}(v) \oplus V \rightarrow \mathbb{F}, \quad (\lambda v, x) \mapsto \lambda.$$

将 φ 的定义域扩张作 V , 得 $\tilde{\varphi} \in \text{ann}(V)$. 由假定, $\tilde{\varphi}(v) = \varphi(v) = 0$, 矛盾.

(E1) 基本逻辑: $\bigcap \cap \bigcap = \bigcap$.

(E2) $\supseteq$ 是基本逻辑. 若 HB, 任取 $f \in \text{ann}(V_1 \cap V_2)$, 则 f 视同商空间 $\frac{V}{V_1 \cap V_2}$ 出发的映射. 不妨设 $V_1 \cap V_2 = 0$, 取

$$f_1 : V_1 \oplus V_2 \rightarrow \mathbb{F}, \quad (x, y) \mapsto (f(x), 0).$$

因此, $f = (f - \widetilde{f_1}) + \widetilde{f_1}$ 属于 $\text{ann}(V_1) + \text{ann}(V_2)$.

(E3) 同 (E1).

(E4) FD, G 见上. G 反例: V 为 $\mathbb{R}$ 上连续函数, $B_1 = \text{span}(\{\delta_r\}_{r \in \mathbb{Q}})$, $B_2 = \text{span}(\{\delta_r\}_{r \in \mathbb{Q}^c})$.

(Z1) 先证明 $\ker \varphi^* \subseteq \text{ann}(\text{im } \varphi)$. 任取 $f \in \ker \varphi^*$, 则 $f \circ \varphi = 0$, 此时 $f \in \text{im } \varphi$. 再证明 $\ker \varphi^* \supseteq \text{ann}(\text{im } \varphi)$. 任取 $f \in \text{ann}(\text{im } \varphi)$, 总有 $f \circ \varphi^* = 0$. 因此 $f \in \ker \varphi^*$.

(Z2) 先证明 $\ker \varphi \subseteq \ker(\text{im } \varphi^*)$. 任意 $x \in \ker \varphi$, 被形如 $f \circ \varphi$ 的映射零化, 即 $x \in \ker(\text{im } \varphi)$. 再证明 HB 下 $\ker \varphi \supseteq \ker(\text{im } \varphi^*)$. 任取 $x \in \ker(\text{im } \varphi^*)$, 若 $\varphi(x) \neq 0$, HB 表明存在 f 使得 $f(\varphi(x)) \neq 0$, 这与 $x \in \ker(\text{im } \varphi^*)$ 矛盾. 因此 $x \in \ker \varphi$.

(Z3) 先证明 $\text{im } \varphi^* \subseteq \text{ann}(\ker \varphi)$. 任取 $f \circ \varphi \in \text{im } \varphi^*$, 则 $f \circ \varphi : \ker \varphi \rightarrow 0$, 因此 $f \circ \varphi \in \text{ann}(\ker \varphi)$. 再证明 HB 下有 $\text{im } \varphi^* \supseteq \text{ann}(\ker \varphi)$. 任取 $f \in \text{ann}(\ker \varphi)$, 则 $\ker f \supseteq \ker \varphi$. 依照 HB, $f' : \text{dom}(\varphi) / \ker \varphi \rightarrow \mathbb{F}$ 通过单射 $\text{dom}(\varphi) / \ker \varphi \hookrightarrow \text{codom}(\varphi)$ 分解. 这说明 f 通过 φ 分解, 也就是 $f \in \text{im } \varphi^*$.

(Z3') 若 φ 是满的, 则 $\text{dom}(\varphi) / \ker \varphi \hookrightarrow \text{codom}(\varphi)$ 是同构. 此时无需 HB.

(Z4) 先证明 $\text{im } \varphi \subseteq \ker(\ker \varphi^*)$. 任取 $\varphi(x) \in \text{im } \varphi$, 则对任意 $f \in \ker \varphi^*$ 都有 $f(\varphi(x)) = 0$, 从而 $\varphi(x) \in \ker(\ker \varphi^*)$. 再证明 HB 下 $\text{im } \varphi \supseteq \ker(\ker \varphi^*)$. 任取 $x \in \ker(\ker \varphi^*)$, 则一切满足 $f \circ \varphi = 0$ 的 f 零化 x . 若 $x \notin \text{im } \varphi$, 考虑映射

$$g : \text{im } \varphi \oplus \text{span}(x) \rightarrow \mathbb{F}, \quad (v, \lambda x) \mapsto \lambda.$$

将上述映射的定义域扩张至全空间, 则得到 $\tilde{g} \circ \varphi = 0$, 但 $\tilde{g}(x) \neq g(x)$, 矛盾.

(L1) 考虑包含映射 $i : S \hookrightarrow V$ 的对偶 i^* , 得 $\text{im}(i^*) \simeq V^* / \ker i^*$. 此处 $\ker i^*$ 就是 $\text{ann}(S)$. 通常情形下, $\text{im}(i^*) \hookrightarrow S^*$ 是单射. 若承认 HB, 则所有 S 上定义的映射可以扩张至 V 上, 也就是 $\text{im}(i^*) \hookrightarrow S^*$ 取等.

(L2) 再证明 $(V/S)^* \simeq \text{ann}(S)$. 自然的商映射 $\pi : V \twoheadrightarrow V/S$ 给出单射 $\pi^* : (V/S)^* \hookrightarrow V^*$. 依商空间的特性, $f \in \text{im } \pi^*$ 当且仅当 f 限制在 S 上是 0, 从而 $f \in \text{ann}(S)$.

Remark. When the general case (G) coincides with the HB case, the proposition is provable without additional axioms. When the FD case differs from the G case, the FD part is proved using dimension arguments. Nonexamples disprove the HB part of statements G1 and E4.

- G1–G2 come from Galois connection,
- E1–E4 are elementary,
- Z1–Z6 are taken from the [textbook of 张贤科](#),
- L1–L2 are taken from the [textbook of 黎景辉等](#).